

# SUPPLEMENT B TO "ON THE RATE OF CONVERGENCE OF FULLY CONNECTED DEEP NEURAL NETWORK REGRESSION ESTIMATES"

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The supplement contains auxiliary results and further proofs of all lemmata, that follow in a straightforward modification from earlier results. To give a complete understanding of all results we provide these lemmata and proofs below.

## B. APPENDIX: AUXILIARY RESULTS AND FURTHER PROOFS.

B.1. *An auxiliary result from the empirical process theory.* In the proof of Theorem 1 we use the following bound on the expected  $L_2$ -error of the least squares estimators.

LEMMA 18. *Assume that the distribution of  $(\mathbf{X}, Y)$  satisfies*

$$\mathbf{E}\{\exp(c_1 \cdot Y^2)\} < \infty$$

*for some constant  $c_1 > 0$  and that the regression function  $m$  is bounded in absolute value. Let  $\tilde{m}_n$  be the least squares estimator*

$$\tilde{m}_n(\cdot) = \arg \min_{f \in \mathcal{F}_n} \frac{1}{n} \sum_{i=1}^n |Y_i - f(\mathbf{X}_i)|^2$$

*based on some function space  $\mathcal{F}_n$  and set  $m_n = T_{c_{53} \cdot \log(n)} \tilde{m}_n$  for some constant  $c_{53} > 0$ . Then  $m_n$  satisfies*

$$\begin{aligned} & \mathbf{E} \int |m_n(\mathbf{x}) - m(\mathbf{x})|^2 \mathbf{P}_{\mathbf{X}}(d\mathbf{x}) \\ & \leq \frac{c_{54} \cdot (\log n)^2 \cdot \sup_{\mathbf{x}_1^n \in (\mathbb{R}^d)^n} \left( \log \left( \mathcal{N}_1 \left( \frac{1}{n \cdot c_{53} \log(n)}, T_{c_{53} \log(n)} \mathcal{F}_n, \mathbf{x}_1^n \right) \right) + 1 \right)}{n} \\ & \quad + 2 \cdot \inf_{f \in \mathcal{F}_n} \int |f(\mathbf{x}) - m(\mathbf{x})|^2 \mathbf{P}_{\mathbf{X}}(d\mathbf{x}) \end{aligned}$$

*for  $n > 1$  and some constant  $c_{54} > 0$ , which does not depend on  $n$  or the parameters in the estimate.*

PROOF. This result follows in a straightforward way from the proof of Theorem 1 in [1] (cf., Supplement of [3]).  $\square$

B.2. *A bound on the covering number.* If the function class  $\mathcal{F}_n$  in Lemma 18 forms a class of fully connected neural networks  $\mathcal{F}(L, r)$ , the following result will help to bound the covering number:

LEMMA 19. *Let  $1/n^{c_{55}} \leq \epsilon < c_{53} \cdot \log(n)/8$  and let  $\mathcal{F}(L, r)$  defined as in (2) where  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  with  $\sigma(x) = \max\{x, 0\}$  and certain constants  $c_{53}, c_{55} > 0$ . Let  $L, r \in \mathbb{N}$ . Then*

$$\log(\mathcal{N}_1(\epsilon, T_{c_{53} \log(n)} \mathcal{F}(L, r), \mathbf{x}_1^n)) \leq c_{56} \cdot \log(n) \cdot \log(L \cdot r^2) \cdot L^2 \cdot r^2$$

*holds for sufficiently large  $n$ ,  $x_1, \dots, x_n \in \mathbb{R}^d$  and a constant  $c_{56} > 0$  independent of  $n$ ,  $L$  and  $r$ .*

PROOF. Due to the fact that all functions  $f(\mathbf{x}) \in T_{c_{53} \log(n)} \mathcal{F}(L, r)$  are bounded by  $c_{53} \log(n)$  and that  $0 < \epsilon < c_{53} \cdot \log(n)/8$  we can apply Lemma 9.2 and Theorem 9.4 in [4] to bound

$$\begin{aligned} & \mathcal{N}_1(\epsilon, T_{c_{53} \log(n)} \mathcal{F}(L, r), \mathbf{x}_1^n) \\ & \leq 3 \left( \frac{4 \cdot e \cdot c_{53} \cdot \log(n)}{\epsilon} \log \left( \frac{6 \cdot e \cdot c_{53} \cdot \log(n)}{\epsilon} \right) \right)^{V_{T_{c_{53} \log(n)} \mathcal{F}(L, r)}^+}. \end{aligned}$$

Theorem 6 in [2] helps us to bound the VC-Dimension (see Definition 1 in [2]) by

$$V_{T_{c_{53} \log(n)} \mathcal{F}(L, r)}^+ \leq V_{\mathcal{F}(L, r)}^+ \leq c_{53} \cdot W \cdot L \cdot \log(W)$$

where  $W$  denotes the total number of weights in the network and  $c_{53} > 0$  is a constant. A fully connected neural network with  $L$  hidden layers and  $r$  neurons per layer consists of

$$\begin{aligned} W &= (d+1) \cdot r + (L-1) \cdot (r+1) \cdot r + r + 1 \\ &= (d+1) \cdot r + L \cdot (r^2 + r) - r^2 + 1 \end{aligned}$$

weights in total. This leads to

$$V_{T_{c_{53} \log(n)} \mathcal{F}(L, r)}^+ \leq c_{54} \cdot L^2 \cdot r^2 \cdot \log(L \cdot r^2)$$

for constants  $c_{54} > 0$  sufficiently large. Combining this with  $\epsilon > 1/n^{c_{55}}$  implies

$$\begin{aligned} & \mathcal{N}_1(\epsilon, T_{c_{53} \log(n)} \mathcal{F}(L, r), \mathbf{x}_1^n) \\ & \leq 3 (4 \cdot e \cdot c_{53} \cdot \log(n) \cdot n^{c_{55}} \cdot \log(6 \cdot e \cdot c_{53} \cdot \log(n) \cdot n^{c_{55}}))^{c_{54} \cdot L^2 \cdot r^2 \cdot \log(L \cdot r^2)} \end{aligned}$$

and therefore

$$\log(\mathcal{N}_1(\epsilon, T_{c_{53} \log(n)} \mathcal{F}(L, r), \mathbf{x}_1^n)) \leq c_{55} \cdot L^2 \cdot r^2 \cdot \log(L \cdot r^2) \cdot \log(n),$$

which shows the assertion.  $\square$

B.3. *Further proofs.* The following lemma presents a neural network, that approximates the square function. This network is essential to build neural networks for more complex tasks.

LEMMA 20. *Let  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  be the ReLU activation function  $\sigma(x) = \max\{x, 0\}$ . Then for any  $R \in \mathbb{N}$  and any  $a \geq 1$  a neural network*

$$\hat{f}_{sq} \in \mathcal{F}(R, 9)$$

*exists such that*

$$\left| \hat{f}_{sq}(x) - x^2 \right| \leq a^2 \cdot 4^{-R}$$

*holds for  $x \in [-a, a]$ .*

PROOF. This proof follows as a slight modification from the proof of Proposition 2 in [5]. For sake of completeness we provide nevertheless a complete proof below. We consider the "tooth" function  $g : [0, 1] \rightarrow [0, 1]$

$$g(x) = \begin{cases} 2x & , x \leq \frac{1}{2} \\ 2 \cdot (1 - x) & , x > \frac{1}{2} \end{cases}$$

and the iterated function

$$g_s(x) = \underbrace{g \circ g \circ \dots \circ g}_s(x).$$

In a *first step of the proof* we show by induction that

$$g_s(x) = \begin{cases} 2^s \left( x - \frac{2k}{2^s} \right) & , x \in \left[ \frac{2k}{2^s}, \frac{2k+1}{2^s} \right], k = 0, 1, \dots, 2^{s-1} - 1 \\ 2^s \left( \frac{2k}{2^s} - x \right) & , x \in \left[ \frac{2k-1}{2^s}, \frac{2k}{2^s} \right], k = 1, 2, \dots, 2^{s-1} \end{cases}.$$

For  $s = 1$  this follows directly from the definition of  $g$  and  $g_1$ . For the induction step we remark that  $(g_s \circ g)(x) = g_s(2x)$  whenever  $x \in [0, \frac{1}{2}]$  and that  $g(x) = g(1 - x)$ . This combined with the symmetry of  $g_s$  (by the inductive hypothesis) implies that for every  $x \in [0, \frac{1}{2}]$

$$\begin{aligned} g_{s+1}(x) &= g_s(g(x)) = g_s(2x) = g_s(1 - 2x) = g_s\left(2 \cdot \left(\frac{1}{2} - x\right)\right) \\ &= g_s\left(g\left(\frac{1}{2} - x\right)\right) = g_s\left(g\left(x + \frac{1}{2}\right)\right) = g_{s+1}\left(x + \frac{1}{2}\right). \end{aligned}$$

Consequently it suffices to consider  $x \in [0, \frac{1}{2}]$  which means

$$(g_s \circ g)(x) = g_s(2x)$$

and together with the inductive hypothesis we have

$$\begin{aligned} (g_s \circ g)(x) &= \begin{cases} 2^s \cdot (2x - \frac{2k}{2^s}) & , 2x \in [\frac{2k}{2^s}, \frac{2k+1}{2^s}], k = 0, 1, \dots, 2^{s-1} - 1 \\ 2^s \cdot (\frac{2k}{2^s} - 2x) & , 2x \in [\frac{2k-1}{2^s}, \frac{2k}{2^s}], k = 1, 2, \dots, 2^{s-1} \end{cases} \\ &= \begin{cases} 2^{s+1} \cdot (x - \frac{2k}{2^{s+1}}) & , x \in [\frac{2k}{2^{s+1}}, \frac{2k+1}{2^{s+1}}], k = 0, 1, \dots, 2^s - 1 \\ 2^{s+1} \cdot (\frac{2k}{2^{s+1}} - x) & , x \in [\frac{2k-1}{2^{s+1}}, \frac{2k}{2^{s+1}}], k = 1, 2, \dots, 2^s, \end{cases} \end{aligned}$$

which shows the assertion.

In a *second step of the proof* we show that the function  $f(x) = x^2$ ,  $x \in [0, 1]$  can be approximated by linear combinations of functions  $g_s$ . Let  $S_R$  be a piecewise linear interpolation of  $f$  with  $2^R + 1$  uniformly distributed break-points  $\frac{k}{2^R}$ ,  $k = 0, \dots, 2^R$

$$S_R\left(\frac{k}{2^R}\right) = \left(\frac{k}{2^R}\right)^2.$$

To determine the error of that piecewise linear interpolation we define the function

$$F(z) = f(z) - S_R(z) + \frac{S_R(x) - f(x)}{(x - \frac{k}{2^R})(x - \frac{k+1}{2^R})} \cdot (z - \frac{k}{2^R})(z - \frac{k+1}{2^R})$$

for  $x \in [\frac{k}{2^R}, \frac{k+1}{2^R}]$  and  $k = 0, \dots, 2^R - 1$ .

We note that  $F(\frac{k}{2^R}) = 0$ ,  $F(\frac{k+1}{2^R}) = 0$  and  $F(x) = 0$ . According to Rolle's theorem, there must be a point  $z_1$ , where  $\frac{k}{2^R} < z_1 < x$  and  $F'(z_1) = 0$  and there must be a point  $z_2$ , where  $x < z_2 < \frac{k+1}{2^R}$  and  $F'(z_2) = 0$ . Using Rolle's theorem again, there must be a point  $\eta$  where  $z_1 < \eta < z_2$  and  $F''(\eta) = 0$ . Thus we get for some  $x \in [\frac{k}{2^R}, \frac{k+1}{2^R}]$

$$\begin{aligned} |f(x) - S_R(x)| &= \left| -\frac{f''(\eta)}{2} \cdot (x - \frac{k}{2^R})(x - \frac{k+1}{2^R}) \right| \\ &\leq \left| (x - \frac{k}{2^R})(x - \frac{k+1}{2^R}) \right| \leq 2^{-2R-2}, \end{aligned}$$

where the last inequality follows since the maximum of

$$h(x) := (x - \frac{k}{2^R})(\frac{k+1}{2^R} - x)$$

is given by  $x = \frac{k}{2^R} + \frac{1}{2} \cdot \frac{1}{2^R}$ .

Furthermore refining the interpolation from  $S_{R-1}$  to  $S_R$  amounts to adjusting

it by a function proportional to a sawtooth function:

$$S_{R-1}(x) - S_R(x) = \frac{g_R(x)}{2^{2R}}.$$

This follows for some  $x \in [\frac{k}{2^{R-1}}, \frac{k+1}{2^{R-1}}]$  ( $k \in \{0, \dots, 2^{R-1} - 1\}$ ), since

$$\begin{aligned} S_{R-1}(x) &= S_{R-1}\left(\frac{k}{2^{R-1}}\right) + \frac{S_{R-1}(\frac{k+1}{2^{R-1}}) - S_{R-1}(\frac{k}{2^{R-1}})}{\frac{1}{2^{R-1}}} \cdot \left(x - \frac{k}{2^{R-1}}\right) \\ &= \left(\frac{k}{2^{R-1}}\right)^2 + \left(\frac{2k+1}{2^{R-1}}\right) \left(x - \frac{k}{2^{R-1}}\right) \end{aligned}$$

and

$$\begin{aligned} S_R(x) &= \begin{cases} S_R\left(\frac{k}{2^{R-1}}\right) + \frac{S_R(\frac{k}{2^{R-1}} + \frac{1}{2^R}) - S_R(\frac{k}{2^{R-1}})}{\frac{1}{2^R}} \cdot \left(x - \frac{k}{2^{R-1}}\right), \\ \quad \text{if } x \in [\frac{k}{2^{R-1}}, \frac{k}{2^{R-1}} + \frac{1}{2^R}] \\ S_R\left(\frac{k}{2^{R-1}} + \frac{1}{2^R}\right) + \frac{S_R(\frac{k+1}{2^{R-1}}) - S_R(\frac{k}{2^{R-1}} + \frac{1}{2^R})}{\frac{1}{2^R}} \\ \quad \cdot \left(x - \frac{k}{2^{R-1}} - \frac{1}{2^R}\right), \\ \quad \text{if } x \in [\frac{k}{2^{R-1}} + \frac{1}{2^R}, \frac{k+1}{2^{R-1}}] \end{cases} \\ &= \begin{cases} \left(\frac{k}{2^{R-1}}\right)^2 + \left(\frac{2k}{2^{R-1}} + \frac{1}{2^R}\right) \left(x - \frac{k}{2^{R-1}}\right), & \text{if } x \in [\frac{k}{2^{R-1}}, \frac{k}{2^{R-1}} + \frac{1}{2^R}] \\ \left(\frac{k}{2^{R-1}}\right)^2 - \frac{2}{2^{2R}} + \left(\frac{4k+3}{2^R}\right) \left(x - \frac{k}{2^{R-1}}\right), & \text{if } x \in [\frac{k}{2^{R-1}} + \frac{1}{2^R}, \frac{k+1}{2^{R-1}}] \end{cases} \end{aligned}$$

Since  $S_0(x) = x$  we can recursively conclude that

$$S_R(x) = x - \sum_{s=1}^R \frac{g_s(x)}{2^{2s}}$$

with

$$|S_R(x) - x^2| \leq 2^{-2R-2}$$

for  $x \in [0, 1]$ .

In a *third step of the proof* we show, that there exists a feedforward neural network that computes  $S_R(x)$  for  $x \in [0, 1]$ . The function  $g(x)$  can be implemented by the network

$$\hat{f}_g(x) = 2 \cdot \sigma(x) - 4 \cdot \sigma\left(x - \frac{1}{2}\right) + 2 \cdot \sigma(x - 1)$$

and the function  $g_s(x)$  can be implemented by a network

$$\hat{f}_{g_s} \in \mathcal{F}(s, 3)$$

with

$$\hat{f}_{g_s}(x) = \underbrace{\hat{f}_g(\hat{f}_g(\dots(\hat{f}_g(x))))}_s.$$

Let

$$\hat{f}_{id}(z) = \sigma(z) - \sigma(-z)$$

with

$$\begin{aligned} \hat{f}_{id}^0(z) &= z & (z \in \mathbb{R}) \\ \hat{f}_{id}^{t+1}(z) &= \hat{f}_{id}(\hat{f}_{id}^t(z)) & (z \in \mathbb{R}, t \in \mathbb{N}_0) \end{aligned}$$

be the network satisfying

$$\hat{f}_{id}^t(z) = z.$$

By combining the networks above we can implement the function  $S_R(x)$  by a network

$$\hat{f}_{sq[0,1]} \in \mathcal{F}(R, 7)$$

recursively defined as follows: We set  $\hat{f}_{1,0}(x) = \hat{f}_{2,0}(x) = x$  and  $\hat{f}_{3,0}(x) = 0$ . Then we set

$$\begin{aligned} \hat{f}_{1,i+1}(x) &= \hat{f}_{id}(\hat{f}_{1,i}(x)), \\ \hat{f}_{2,i+1}(x) &= \hat{f}_g(\hat{f}_{2,i}(x)) \end{aligned}$$

and

$$\hat{f}_{3,i+1}(x) = \hat{f}_{3,i}(x) - \hat{f}_g(\hat{f}_{2,i}(x))/2^{2(i+1)}$$

for  $i \in \{0, 1, \dots, R-2\}$  and

$$\hat{f}_{sq[0,1]}(x) = \hat{f}_{id}(\hat{f}_{1,R-1}(x) + \hat{f}_{3,R-1}(x) - \hat{f}_g(\hat{f}_{2,i-1}(x))/2^{2R}).$$

This implies

$$\begin{aligned} \hat{f}_{sq[0,1]}(x) &= \hat{f}_{id}^R(x) - \frac{1}{2^{2R}} \hat{f}_{g_R}(x) - \hat{f}_{id} \left( \frac{1}{2^{2(R-1)}} \hat{f}_{g_{R-1}}(x) \right. \\ &\quad \left. - \hat{f}_{id} \left( \frac{1}{2^{2(R-2)}} \hat{f}_{g_{R-2}}(x) - \dots - \hat{f}_{id} \left( \frac{1}{2^2} \hat{f}_{g_1}(x) \right) \right) \right) \\ &= S_R(x), \end{aligned}$$

hence  $\hat{f}_{sq_{[0,1]}}(x)$  satisfies

$$(47) \quad |\hat{f}_{sq_{[0,1]}}(x) - x^2| \leq 2^{-2R-2}$$

for  $x \in [0, 1]$ .

In a *last step of the proof* we show that we can also approximate the function  $f(x) = x^2$  by a neural network, if  $x \in [-a, a]$ . Therefore let  $f_{tran} : [-a, a] \rightarrow [0, 1]$  with

$$f_{tran}(z) = \frac{z}{2a} + \frac{1}{2}$$

be the function that transfers the value of  $x \in [-a, a]$  in the interval, where (47) holds. Set

$$\hat{f}_{sq}(x) = 4a^2 \hat{f}_{sq_{[0,1]}}(f_{tran}(x)) - 2a \cdot \hat{f}_{id}^R(x) - a^2.$$

Since

$$x^2 = 4a^2 \cdot \left( \frac{x}{2a} + \frac{1}{2} \right)^2 - 2ax - a^2$$

we have

$$\begin{aligned} & |\hat{f}_{sq}(x) - x^2| \\ & \leq 4a^2 \cdot |\hat{f}_{sq_{[0,1]}}(f_{tran}(x)) - (f_{tran}(x))^2| + 2a|\hat{f}_{id}^R(x) - x| \\ & \leq 4a^2 \cdot 2^{-2R-2} = a^2 \cdot 4^{-R}. \end{aligned}$$

□

PROOF OF LEMMA 4. Let

$$\hat{f}_{sq} \in \mathcal{F}(R, 9)$$

be the neural network from Lemma 4 satisfying

$$|\hat{f}_{sq}(x) - x^2| \leq 4 \cdot a^2 \cdot 4^{-R}$$

for  $x \in [-2a, 2a]$ , and set

$$\hat{f}_{mult}(x, y) = \frac{1}{4} \cdot \left( \hat{f}_{sq}(x + y) - \hat{f}_{sq}(x - y) \right).$$

Since

$$x \cdot y = \frac{1}{4} \left( (x + y)^2 - (x - y)^2 \right)$$

we have

$$\begin{aligned}
|\hat{f}_{mult}(x, y) - x \cdot y| &\leq \frac{1}{4} \cdot \left| \hat{f}_{sq}(x + y) - (x + y)^2 \right| + \frac{1}{4} \cdot \left| (x - y)^2 - \hat{f}_{sq}(x - y) \right| \\
&\leq \frac{1}{4} \cdot 2 \cdot 4 \cdot a^2 \cdot 4^{-R} \\
&\leq 2 \cdot a^2 \cdot 4^{-R}
\end{aligned}$$

for  $x, y \in [-a, a]$ . □

PROOF OF LEMMA 8. We set  $q = \lceil \log_2(d) \rceil$ . The feedforward neural network  $\hat{f}_{mult,d}$  with  $L = R \cdot q$  hidden layers and  $r = 18d$  neurons in each layer is constructed as follows: Set

$$(48) \quad (z_1, \dots, z_{2^q}) = \left( x^{(1)}, x^{(2)}, \dots, x^{(d)}, \underbrace{1, \dots, 1}_{2^q - d} \right).$$

In the construction of our network we will use the network  $\hat{f}_{mult}$  of Lemma 6, which satisfies

$$(49) \quad |\hat{f}_{mult}(x, y) - x \cdot y| \leq 2 \cdot (4^{N+1} a^{N+1})^2 \cdot 4^{-R}$$

for  $x, y \in [-4^d a^d, 4^d a^d]$ . In the first  $R$  layers we compute

$$\hat{f}_{mult}(z_1, z_2), \hat{f}_{mult}(z_3, z_4), \dots, \hat{f}_{mult}(z_{2^q-1}, z_{2^q}),$$

which can be done by  $R$  layers of  $18 \cdot 2^{q-1} \leq 18 \cdot d$  neurons. E.g., in case in case  $z_l = x^{(d)}$  and  $z_{l+1} = 1$  we have

$$\hat{f}_{mult}(z_l, z_{l+1}) = \hat{f}_{mult}(x^{(d)}, 1).$$

As a result of the first  $R$  layers we get a vector of outputs which has length  $2^{q-1}$ . Next we pair these outputs and apply  $\hat{f}_{mult}$  again. This procedure is continued until there is only one output left. Therefore we need  $L = Rq$  hidden layers and at most  $18d$  neurons in each layer.

By (49) and  $R \geq \log_4(2 \cdot 4^{2 \cdot d} \cdot a^{2 \cdot d})$  we get for any  $l \in \{1, \dots, d\}$  and any  $z_1, z_2 \in [-(4^l - 1) \cdot a^l, (4^l - 1) \cdot a^l]$

$$|\hat{f}_{mult}(z_1, z_2)| \leq |z_1 \cdot z_2| + |\hat{f}_{mult}(z_1, z_2) - z_1 \cdot z_2| \leq (4^l - 1)^2 a^{2l} + 1 \leq (4^{2l} - 1) \cdot a^{2l}.$$

From this we get successively that all outputs of layer  $l \in \{1, \dots, q - 1\}$  are contained in the interval  $[-(4^{2^l} - 1) \cdot a^{2^l}, (4^{2^l} - 1) \cdot a^{2^l}]$ , hence in particular



they are contained in the interval  $[-4^d a^d, 4^d a^d]$  where inequality (49) does hold.

Define  $\hat{f}_{2^q}$  recursively by

$$\hat{f}_{2^q}(z_1, \dots, z_{2^q}) = \hat{f}_{mult}(\hat{f}_{2^{q-1}}(z_1, \dots, z_{2^{q-1}}), \hat{f}_{2^{q-1}}(z_{2^{q-1}+1}, \dots, z_{2^q}))$$

and

$$\hat{f}_2(z_1, z_2) = \hat{f}_{mult}(z_1, z_2),$$

and set

$$\Delta_l = \sup_{z_1, \dots, z_{2^l} \in [-a, a]} |\hat{f}_{2^l}(z_1, \dots, z_{2^l}) - \prod_{i=1}^{2^l} z_i|.$$

Then

$$|\hat{f}_{mult, d}(\mathbf{x}) - \prod_{i=1}^d x^{(i)}| \leq \Delta_q$$

and from

$$\Delta_1 \leq 2 \cdot (4^d \cdot a^d)^2 \cdot 4^{-R}$$

(which follows from (49)) and

$$\begin{aligned} \Delta_q &\leq \sup_{z_1, \dots, z_{2^q} \in [-a, a]} |\hat{f}_{mult}(\hat{f}_{2^{q-1}}(z_1, \dots, z_{2^{q-1}}), \hat{f}_{2^{q-1}}(z_{2^{q-1}+1}, \dots, z_{2^q})) \\ &\quad - \hat{f}_{2^{q-1}}(z_1, \dots, z_{2^{q-1}}) \cdot \hat{f}_{2^{q-1}}(z_{2^{q-1}+1}, \dots, z_{2^q})| \\ &+ \sup_{z_1, \dots, z_{2^q} \in [-a, a]} \left| \hat{f}_{2^{q-1}}(z_1, \dots, z_{2^{q-1}}) \cdot \hat{f}_{2^{q-1}}(z_{2^{q-1}+1}, \dots, z_{2^q}) \right. \\ &\quad \left. - \left( \prod_{i=1}^{2^{q-1}} z_i \right) \cdot \hat{f}_{2^{q-1}}(z_{2^{q-1}+1}, \dots, z_{2^q}) \right| \\ &+ \sup_{z_1, \dots, z_{2^q} \in [-a, a]} \left| \left( \prod_{i=1}^{2^{q-1}} z_i \right) \cdot \hat{f}_{2^{q-1}}(z_{2^{q-1}+1}, \dots, z_{2^q}) \right. \\ &\quad \left. - \left( \prod_{i=1}^{2^{q-1}} z_i \right) \cdot \prod_{i=2^{q-1}+1}^{2^q} z_i \right| \\ &\leq 2 \cdot (4^d \cdot a^d)^2 \cdot 4^{-R} + 2 \cdot 4^{2^{q-1}} \cdot a^{2^{q-1}} \cdot \Delta_{q-1} \end{aligned}$$

(where the last inequality follows from (49) and the fact that all outputs of layer  $l \in \{1, \dots, q-1\}$  are contained in the interval  $[-4^{2^l} a^{2^l}, 4^{2^l} a^{2^l}]$ ) we get

for  $x \in [-a, a]^d$

$$\begin{aligned}
& |\hat{f}_{mult,d}(\mathbf{x}) - \prod_{i=1}^d x^{(i)}| \\
& \leq \Delta_q \\
& \leq 2 \cdot (4^d \cdot a^d)^2 \cdot 4^{-R} \cdot 4^{1+2+\dots+2^{q-1}} \cdot a^{1+2+\dots+2^{q-1}} \cdot (1+2+\dots+2^{q-1}) \\
& \leq (4^d \cdot a^d)^2 \cdot 4^{-R} \cdot 4^{2d+1} \cdot a^{2d} \cdot d \\
& = 4 \cdot 4^{4d} \cdot a^{4d} \cdot d \cdot 4^{-R}.
\end{aligned}
\tag{50}$$

□

PROOF OF LEMMA 5. In the *first step of the proof* we will construct a neural network  $\hat{f}_m$ , that approximates

$$y \cdot m(\mathbf{x}) = y \cdot \prod_{k=1}^d \left(x^{(k)}\right)^{r_k}, \quad x \in [-a, a]^d, y \in [-a, a],$$

where  $m \in \mathcal{P}_N$  and  $r_1, \dots, r_d \in \mathbb{N}_0$  with  $r_1 + \dots + r_d \leq N$ . By using  $y$  ones and some of the  $x^{(i)}$  several times, if necessary, Lemma 10 can be extended in a straightforward way to monomials. Here we substitute  $d$  by  $N+1$  and finally can show that a network

$$\hat{f}_m \in \mathcal{F}(R \cdot \lceil \log_2(N+1) \rceil, 18 \cdot (N+1))$$

achieves an approximation error

$$|\hat{f}_m(\mathbf{x}, y) - y \cdot m(\mathbf{x})| = 4 \cdot 4^{4(N+1)} \cdot a^{4(N+1)} \cdot (N+1) \cdot 4^{-R}.$$

(51)

In the *second step* we finish the proof. Let

$$p\left(\mathbf{x}, y_1, \dots, y_{\binom{d+N}{d}}\right) = \sum_{i=1}^{\binom{d+N}{d}} r_i \cdot y_i \cdot m_i(\mathbf{x}).$$

We can conclude

$$\left| p\left(\mathbf{x}, y_1, \dots, y_{\binom{d+N}{d}}\right) - \sum_{i=1}^{\binom{d+N}{d}} r_i \cdot \hat{f}_{m_i}(\mathbf{x}, y_i) \right|$$

$$\begin{aligned}
&\leq \sum_{i=1}^{\binom{d+N}{d}} |r_i| \cdot \left| y_i \cdot m_i(\mathbf{x}) - \hat{f}_{m_i}(\mathbf{x}, y_i) \right| \\
&\leq \binom{d+N}{d} \cdot \bar{r}(p) \cdot 4 \cdot 4^{4 \cdot (N+1)} \cdot a^{4 \cdot (N+1)} \cdot (N+1) \cdot 4^{-R}.
\end{aligned}$$

□

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